

Interaction Terms and Polynomial Regression

CS1090A Introduction to Data Science
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Photo: Amrita Patil

Lecture Outline

Interaction Effects in Regression Models

Polynomial Regression: Extending Linear Models

Model Selection Techniques: Focus on Cross-Validation

Too many predictors and collinearity lead to
OVERFITTING!

Game Time



If your model was a student, what would overfitting be like?

Options:

- A. Studying just the night before the test
- B. Memorizing every lecture, lab and OH word-for-word
- C. Only studying one chapter for all subjects
- D. Taking extensive notes but forgetting to actually understand the concepts

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If your model was a TF, what would overfitting be like?

Options:

- A. Grading papers while wearing 3D glasses to "see the errors in a new dimension"
- B. Using a "Magic 8-Ball" to decide students' grades
- C. Give everyone the first letter that comes on their name. Sorry Frank
- D. Subtract points for every answer that does not include the word overfitting

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Overfitting occurs when a model learns the training data too well, including its noise and outliers, resulting in poor performance on new, unseen data.

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Assumptions of Linear Regression

It is defined as:

$$f(x) = \beta_0 + \beta_1 x$$

This assumes a **linear** relationship

Loss function:

$$L = \frac{1}{n} \sum_{i=1}^n (y_i - \beta_0 + \beta_1 x_i)^2$$

We assume each error contributes **independently** to the loss.

Residuals are assumed to be **normally** distributed.

No weights → all errors are treated as having equal variance (**homoscedasticity**).

Assumptions of Linear Regression

Linearity: Relationship between variables is linear.

$$f(x) = \beta_0 + \beta_1 x$$

Independence: No correlation between errors and predictors.

Homoscedasticity: Constant variance of residuals.

Normality of Residuals: Residuals are normally distributed.

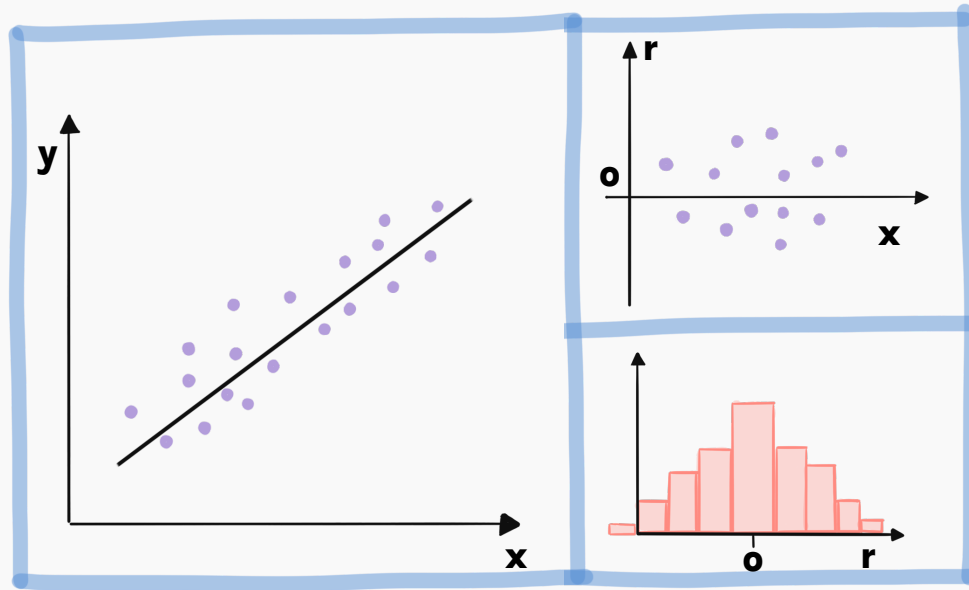
$$\begin{aligned} y &= f(x) + \epsilon \\ \epsilon &\sim N(0, \sigma_\epsilon) \\ L(\beta_0, \beta_1) &= MSE \end{aligned}$$

Other things to consider

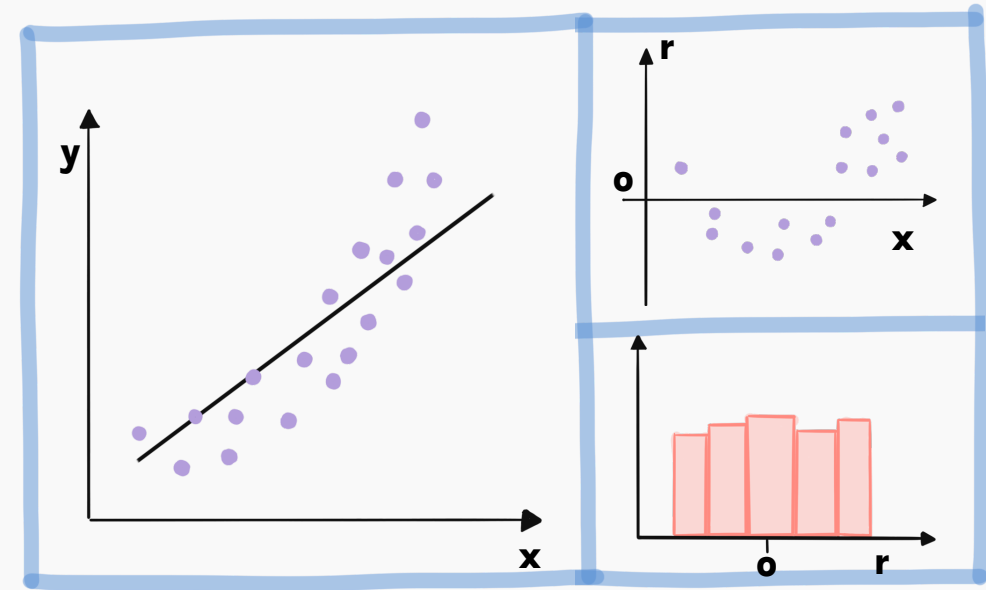
Fixed X: Independent variables are error-free.

No Multicollinearity: Low correlation between predictors.

Residual Analysis



Linear assumption is correct. There is no obvious relationship between residuals and x . Histogram of residuals is **symmetric** and **normally distributed**.



Linear assumption is incorrect. There is an obvious relationship between residuals and x . Histogram of residuals is symmetric but **not normally distributed**.

Note: For multi-regression, we plot the residuals vs predicted y , $\hat{y}$, since there are too many x 's and that could wash out the relationship.

Beyond linearity: synergy effect or interaction effect

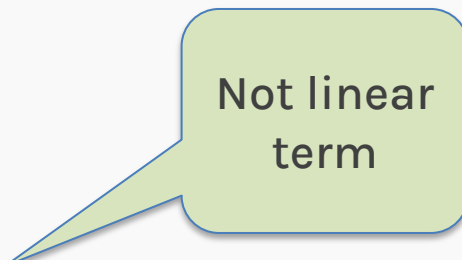
We assumed that the average effect on *sales* of a one-unit increase in *TV*, is always β_1 regardless of the amount spent on *radio* or *newspaper*.

Synergy effect or **interaction effect** states that when an increase on the *radio budget* affects the effectiveness of the *TV* spending on *sales*.

To account for it, we simply add a term as:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2$$



Not linear
term

Beyond linearity: synergy effect or interaction effect

We will demonstrate the effect of the synergy (interaction) term on the **credit card balance dataset** introduced earlier.

This dataset includes variables such as balance, age, number of cards, education, income, credit limit, and rating for a sample of potential customers.

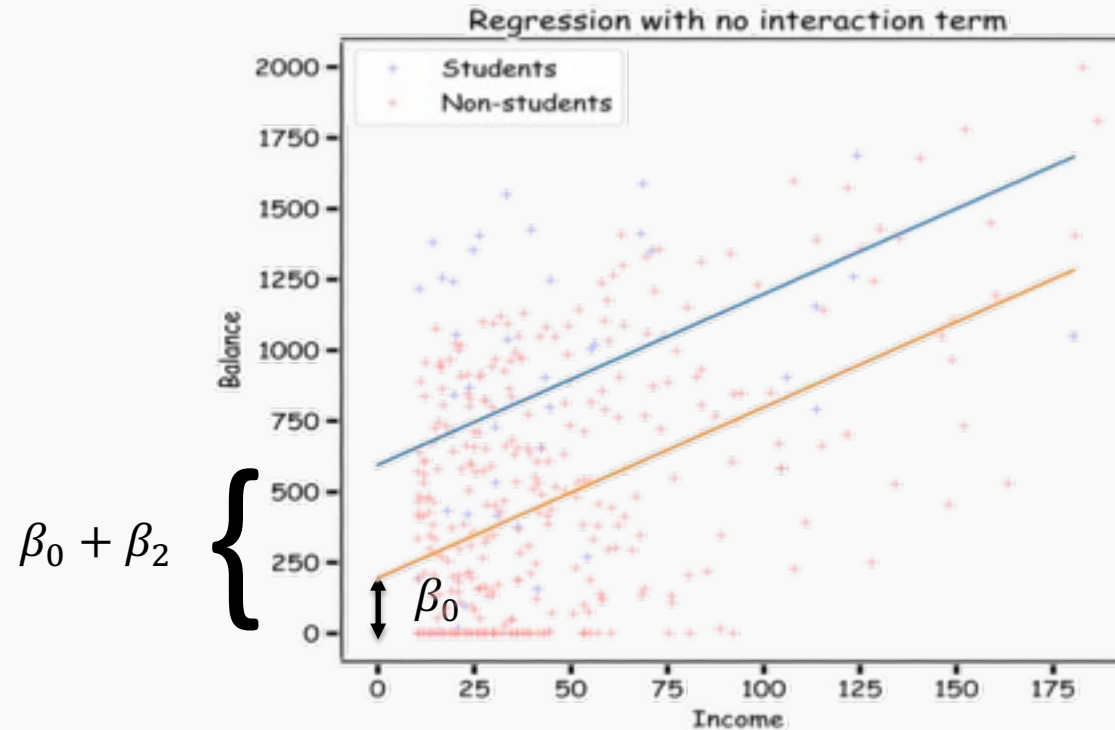
For this example, we focus on just two predictors: **Income** and **Student**.

Income	Limit	Rating	Cards	Age	Education	Sex	Student	Married	Ethnicity	Balance
14.890	3606	283	2	34	11	Male	No	Yes	Caucasian	333
106.02	6645	483	3	82	15	Female	Yes	Yes	Asian	903
104.59	7075	514	4	71	11	Male	No	No	Asian	580
148.92	9504	681	3	36	11	Female	No	No	Asian	964
55.882	4897	357	2	68	16	Male	No	Yes	Caucasian	331

What does it mean? First consider the case without the interaction term

$$\text{balance} = \beta_0 + \beta_1 \times \text{income} + \beta_2 \times \text{student}$$

$$\text{student} = \begin{cases} 0 & \text{balance} = \beta_0 + \beta_1 \times \text{income} \\ 1 & \text{balance} = \beta_0 + \beta_1 \times \text{income} + \beta_2 \end{cases} \rightarrow \text{balance} = \underbrace{(\beta_0 + \beta_2)}_{\text{intercept}} + \underbrace{\beta_1}_{\text{slope}} \times \text{income}$$

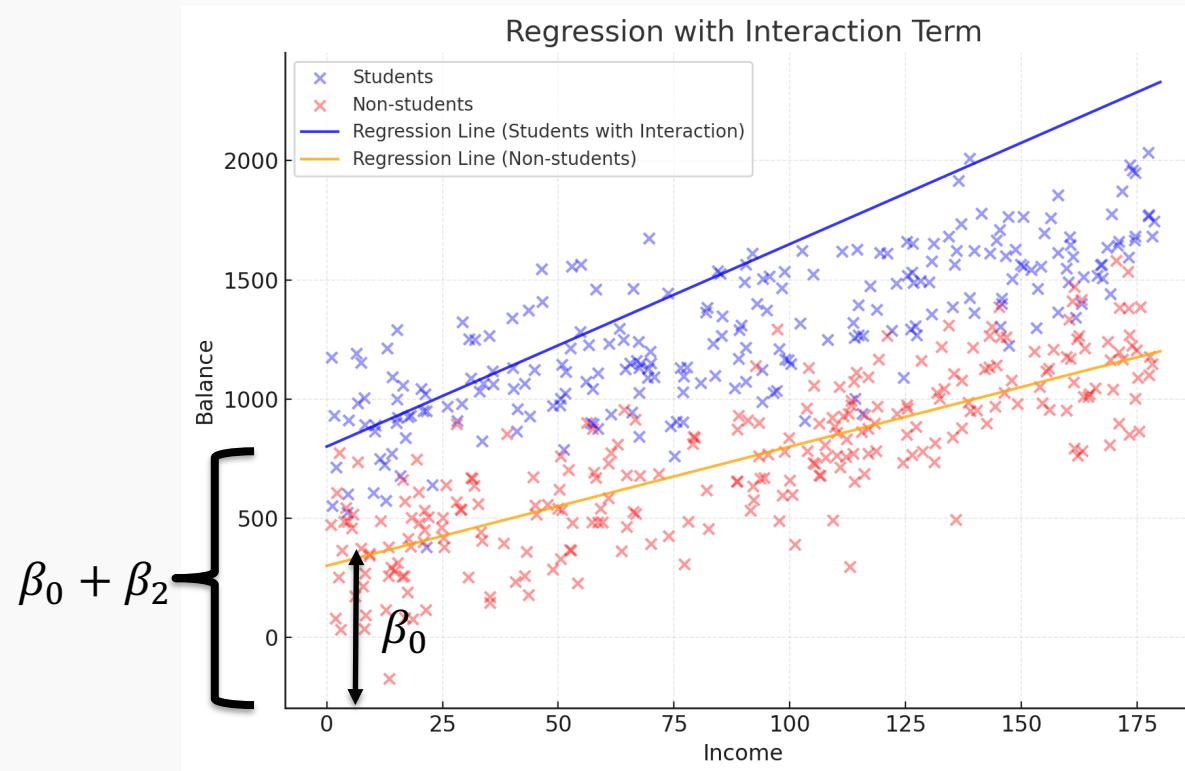


With the interaction term

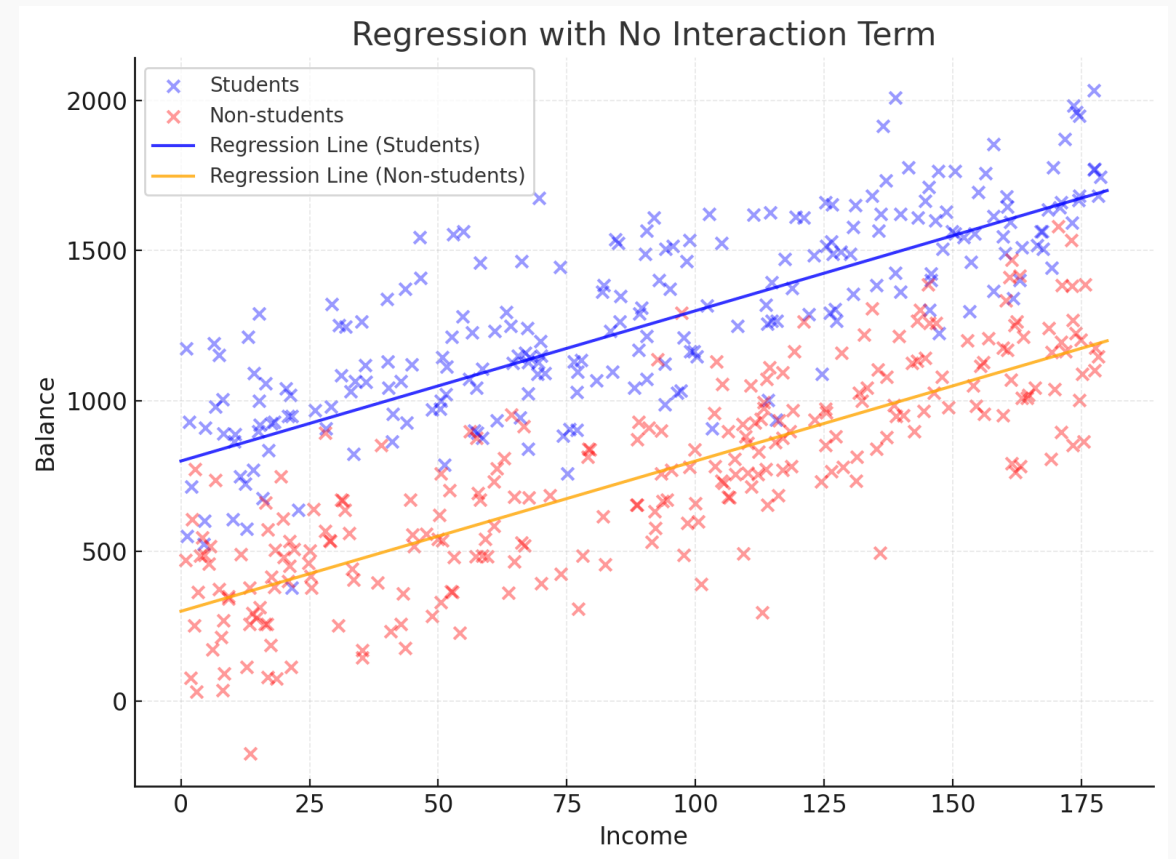
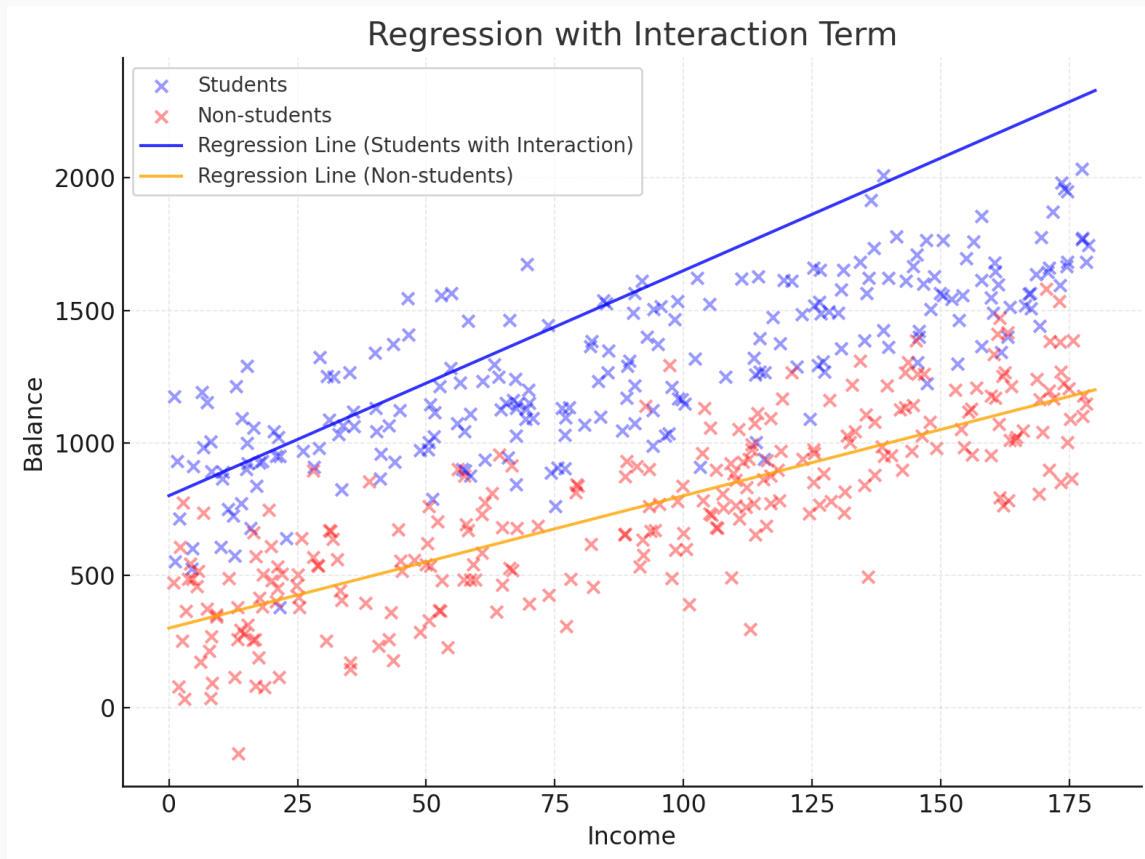
$$\text{balance} = \beta_0 + \beta_1 \times \text{income} + \beta_2 \times \text{student} + \beta_3 \times \text{income} \times \text{student}$$

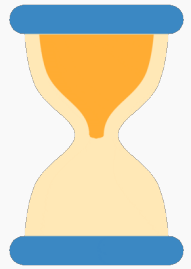
$$\text{student} = \begin{cases} 0 & \text{balance} = \beta_0 + \beta_1 \times \text{income} \\ 1 & \text{balance} = \beta_0 + \beta_1 \times \text{income} + \beta_2 + \beta_3 \times \text{income} \end{cases}$$

$$\rightarrow \text{balance} = \underbrace{(\beta_0 + \beta_2)}_{\text{intercept}} + \underbrace{(\beta_1 + \beta_3)}_{\text{slope}} \times \text{income}$$



With v/s without interaction term





Digestion Time

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Lecture Outline

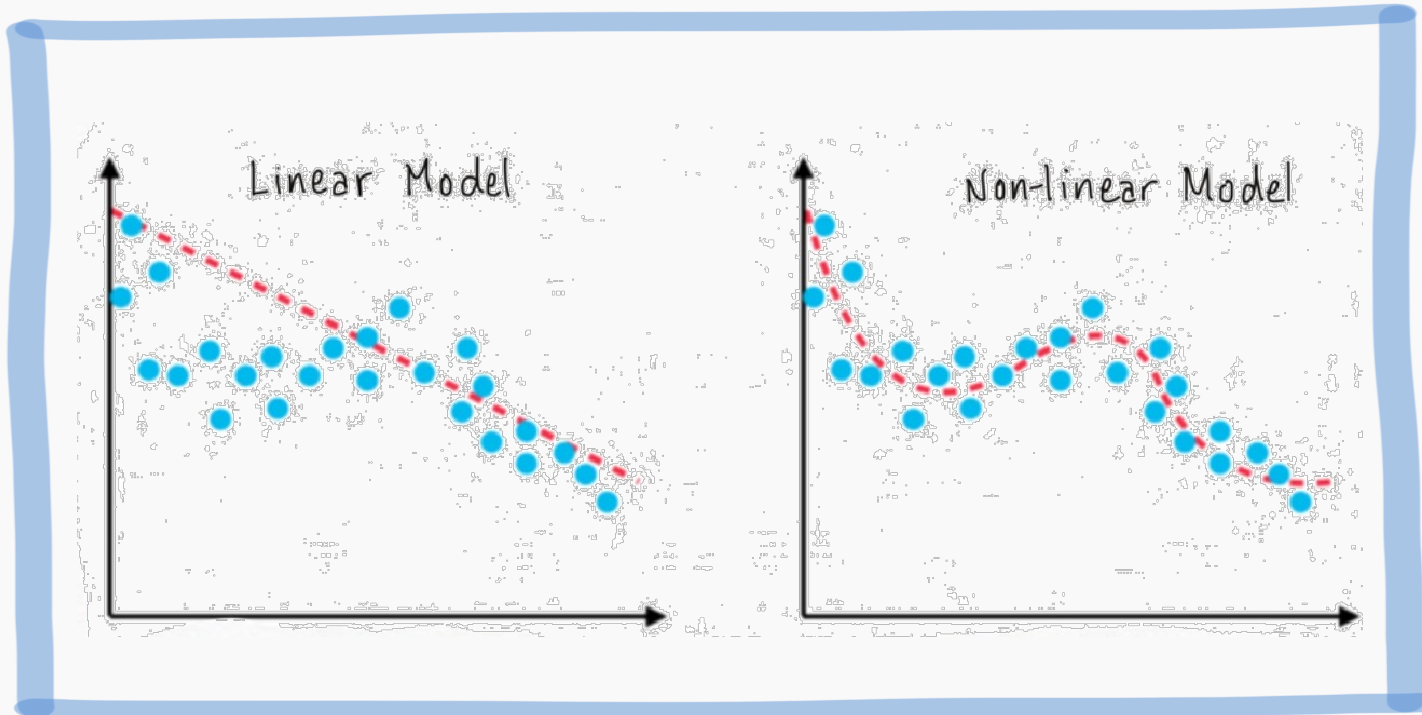
Interaction Effects in Regression Models

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Fitting non-linear data

Multi-linear models can fit large datasets with many predictors. But the relationship between predictor and target isn't always linear.



We want a model:

$$y = f_{\beta}(x) + \epsilon$$

where f is a **non-linear** function and β is a vector of the **parameters** of f .

Polynomial Regression

The **simplest** non-linear model we can consider, for a response Y and a predictor x , is a polynomial model of degree M ,

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \cdots + \beta_M x^M$$

Just as in the case of multi-linear regression, **polynomial** regression is a **special case** of linear regression

HOW?



Polynomial Regression

The design matrix for a polynomial regression would be:

To the
power of
 M

$$\mathbf{Y} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} 1 & x_1^1 & \dots & x_1^M \\ 1 & x_2^1 & \dots & x_2^M \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & \dots & x_n^M \end{pmatrix}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_M \end{pmatrix}.$$

Polynomial Regression

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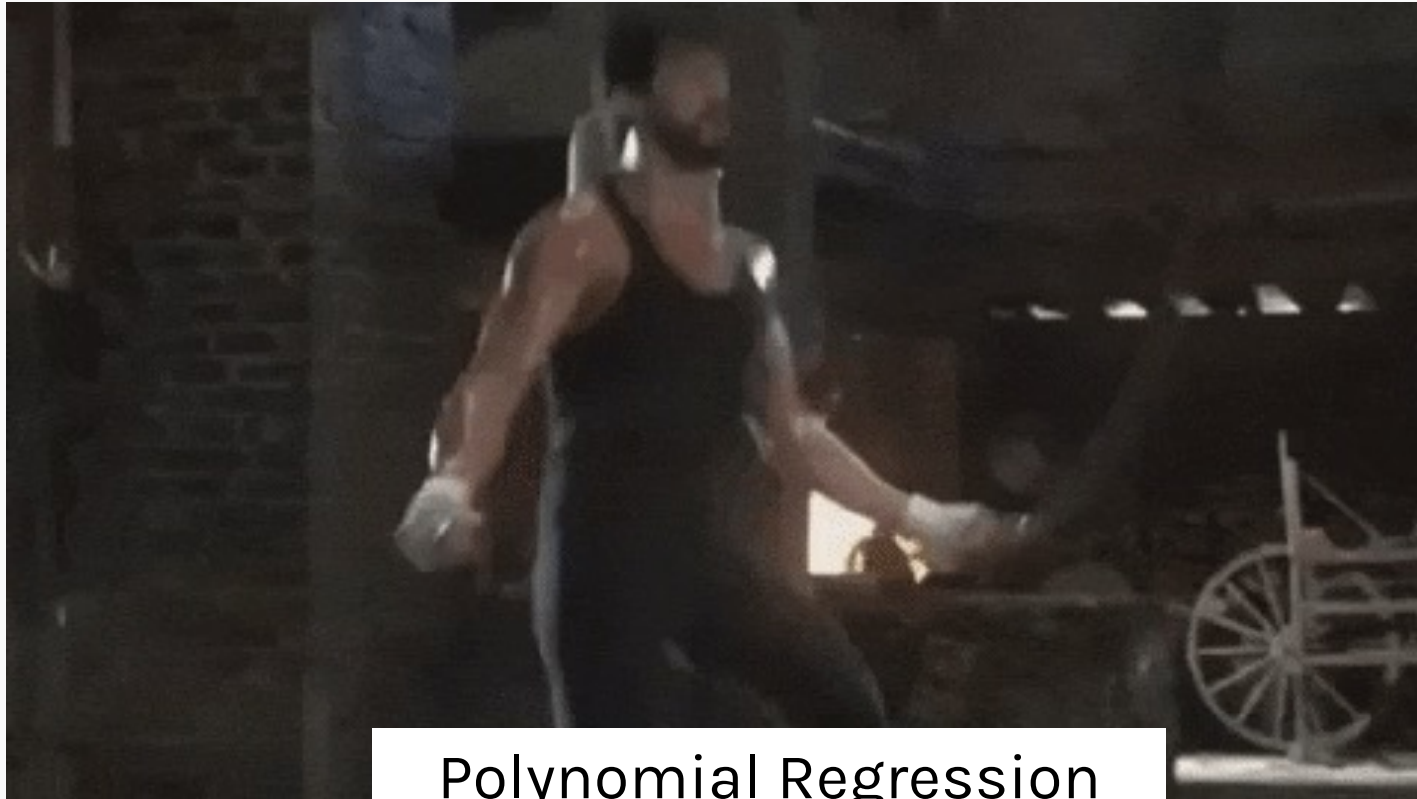
This looks a lot like **multi-linear regression** where the predictors are powers of x!

Multi-Regression

$$\mathbf{Y} = \begin{pmatrix} y_1 \\ \vdots \\ y_y \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} 1 & x_{1,1} & \dots & x_{1,J} \\ 1 & x_{2,1} & \dots & x_{2,J} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n,1} & \dots & x_{n,J} \end{pmatrix}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_J \end{pmatrix},$$

Now, that we know what the model looks like

Let's look at the model training!



Polynomial Regression

Model Training

Give a dataset $\{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$, to find the optimal polynomial model:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \dots + \beta_M x^M$$

1. We **transform** the data by adding new predictors:

$$\tilde{x} = [1, \tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_M]$$

where $\tilde{x}_k = x^k$

2. We find the parameter by **minimizing** the MSE using vector calculus yields, as in multi-linear regression

$$\hat{\beta} = (\tilde{X}^T \tilde{X})^{-1} \tilde{X}^T y$$

We can generate $\tilde{x}$ by calling:

```
sklearn.preprocessing.PolynomialFeatures(degree=?)
```

```
sklearn.linear_model.LinearRegression.fit()
```

Model Training

Give a dataset $\{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$
model:

$$y = \beta_0 + \beta_1 x$$

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$$\tilde{x} = [1, x_1]$$

where $\tilde{x}_k = x^k$

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$$\hat{\beta} = (\tilde{X}^T \tilde{X})^{-1} \tilde{X}^T y$$

`sklearn.linear_model.Linear
Regression.fit()`

We can also perform multi-polynomial regression in the same way.

BUT be careful:

A. Sklearn will include the interaction terms

B. The new design matrix will include the 1 column so no need to fit for intercept

Model Training

Giv
m

ial

Polynomial regression model is linear in terms of the coefficients, not the relationship between the independent and dependent variables

1. y

2.

yields, as in linear regression.

lling:

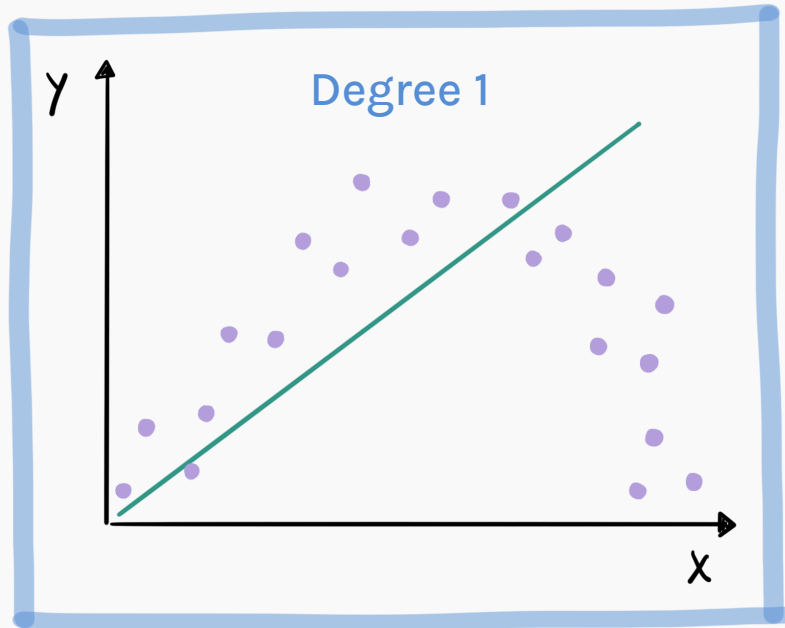
g.Poly
ee=?)

$$\hat{\beta} = (\tilde{X}^T \tilde{X})^{-1} \tilde{X}^T y$$

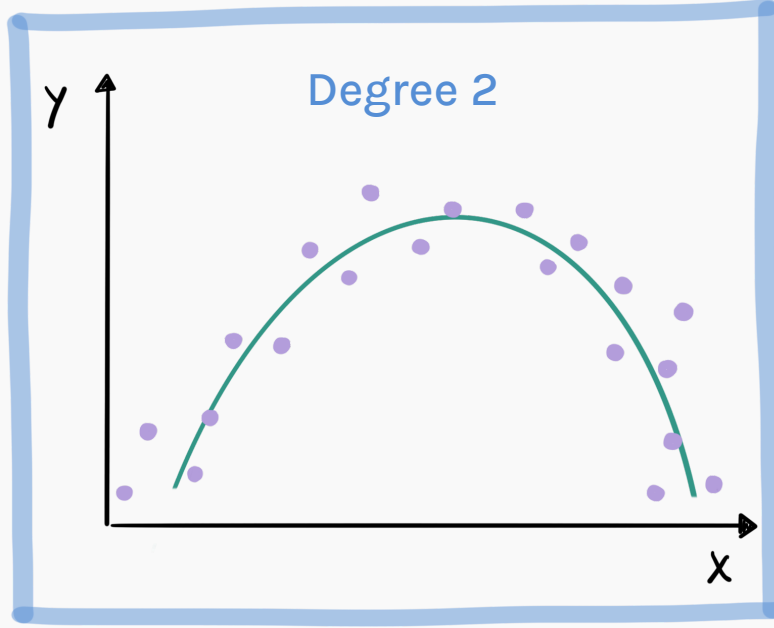
```
sklearn.linear_model.Linear  
Regression.fit()
```

Polynomial Regression (cont.)

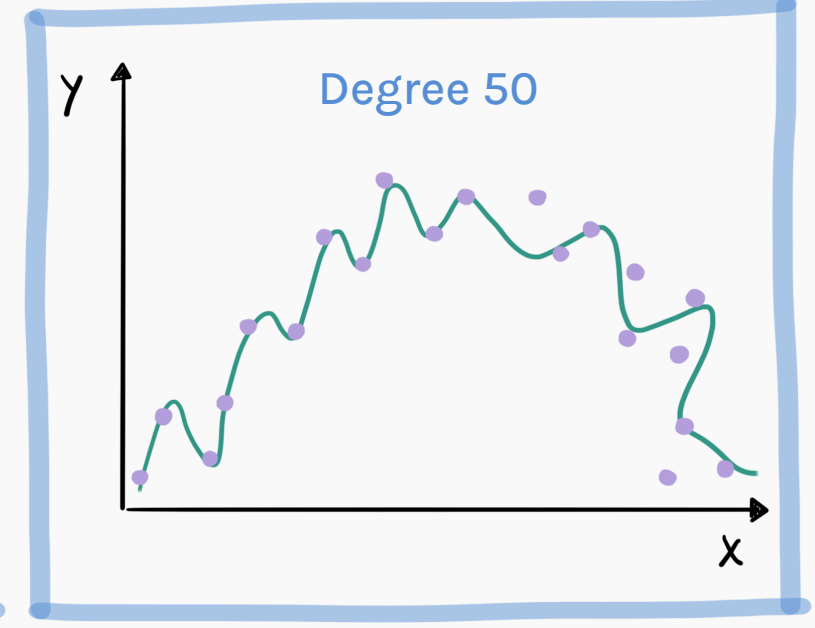
Fitting a polynomial model requires choosing a degree.



Underfitting: when the degree is too low, the model cannot fit the trend.



We want a model that fits the trend and ignores the noise.



Overfitting: when the degree is too high, the model fits all the noisy data points.

Feature Scaling

Do we need to scale out features for polynomial regression?

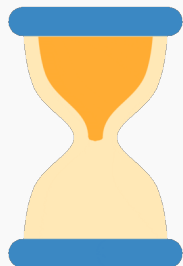
Linear regression, $Y = X\beta$, is **invariant under scaling**. If X is multiplied by some number λ , then β will be scaled by $\frac{1}{\lambda}$ and MSE will be identical.

However, if the range of X is small or large, then we run into troubles. Consider a polynomial degree of 20 and the maximum or minimum value of any predictor is large or small. Those numbers to the 20th power will be **problematic**.

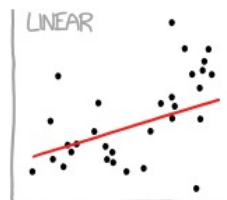
It is always a good idea to **scale** X when considering **polynomial regression**:

$$X^{norm} = \frac{X - \bar{X}}{\sigma_X}$$

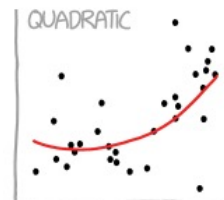
Note: sklearn's `StandardScaler()` can do this.



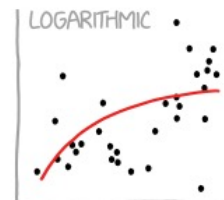
CURVE-FITTING METHODS AND THE MESSAGES THEY SEND



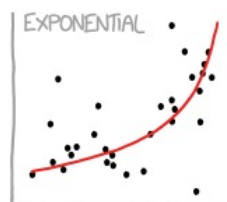
"HEY, I DID A
REGRESSION."



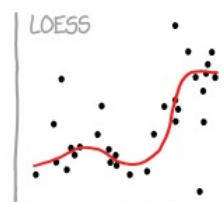
"I WANTED A CURVED
LINE, SO I MADE ONE
WITH MATH."



"LOOK, IT'S
TAPERING OFF!"



"LOOK, IT'S GROWING
UNCONTROLLABLY!"



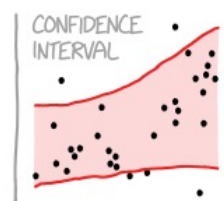
"I'M SOPHISTICATED, NOT
LIKE THOSE BUMBLING
POLYNOMIAL PEOPLE."



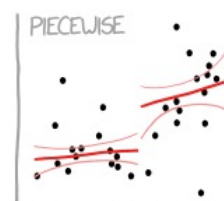
"I'M MAKING A
SCATTER PLOT BUT
I DON'T WANT TO."



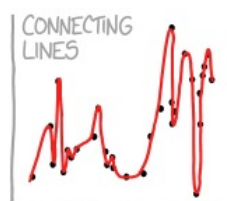
"I NEED TO CONNECT THESE
TWO LINES, BUT MY FIRST IDEA
DIDN'T HAVE ENOUGH MATH."



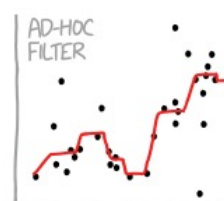
"LISTEN, SCIENCE IS HARD.
BUT I'M A SERIOUS
PERSON DOING MY BEST."



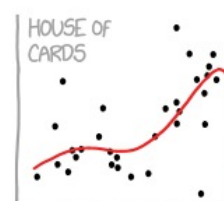
"I HAVE A THEORY,
AND THIS IS THE ONLY
DATA I COULD FIND."



"I CLICKED 'SMOOTH
LINES' IN EXCEL."



"I HAD AN IDEA FOR HOW
TO CLEAN UP THE DATA.
WHAT DO YOU THINK?"



"AS YOU CAN SEE, THIS
MODEL SMOOTHLY FITS
THE- WAIT NO NO DON'T
EXTEND IT AAAAAA!!!"

Too many predictors, collinearity, too many interaction terms and high degree of polynomial leads to

Too many predictors, collinearity, **too many interaction**
terms and high degree of polynomial
leads to
OVERFITTING!



Too many predictors, collinearity, too many interaction terms and high degree of polynomial and model selection leads to

Too many predictors, collinearity, too many interaction terms and high degree of polynomial and model selection leads to
THESE ARE OVERFITTED STUDENTS!

